PRODUCT OPPORTUNITY DISCOVERY

QueueLess

AI-based smart appointment and virtual queue management for hospitals

SUBMITTED BY

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Course Product Management
Project Reducing hospital waiting time through AI-based queue management
Updated October 2026

At a Glance

Problem	Patients wait an hour or more to see a doctor and never know when their turn will come.
Solution	QueueLess: digital tokens, live queue position and an AI-predicted waiting time with delay alerts.
Users	Hospital outpatients, reception staff and doctors.
Key evidence	In Indian OPD studies, patients spend about 85–94% of their visit waiting; consultations average 6–7 minutes.
Success metric	Average actual time spent in the waiting room, plus prediction accuracy.
Next step	Test a Figma prototype with 5 patients and 2 receptionists, then pilot in one hospital department.

1. Problem Statement

Patients in Indian hospitals often wait an hour or more to see a doctor, and in busy departments much longer, even after booking an appointment. Appointment slots are static, nobody tells patients where they are in the queue, and waiting rooms fill up.

> *“Patients struggle with long and unpredictable waiting times in hospitals because appointment systems are static and do not provide real-time queue updates.”*

Why this problem matters

People now expect the same digital convenience from healthcare that they get from food delivery or ride-booking apps, where they always know when something will arrive. Many hospitals still manage queues manually. The common frustrations are:

- Long waiting time, with no estimate of when the consultation will happen
- Crowded waiting rooms and unnecessary time spent at the hospital
- Missed work and lost productive hours
- Repeated enquiries to reception staff

Who is affected

Stakeholder	How the problem affects them
Patients	Lost time, anxiety and uncertainty; difficulty planning the day around a visit
Doctors	Uneven patient flow; idle gaps followed by rushed periods
Reception staff	Constant “how long will it take?” questions; manual token handling
Hospital management	Crowding, lower patient satisfaction and weaker retention

2. Evidence That the Problem Exists

I carried out secondary research by reviewing hospital Google Reviews, healthcare discussion forums and patient-experience discussions. The same three themes appeared repeatedly: long waits despite appointments, no visibility of the queue, and time wasted. The statements below are representative of those themes.

> *"Even after taking an appointment, I had to wait almost two hours."*
>
> *"No one informed me how many patients were ahead of me."*
>
> *"Half of my day was wasted in waiting."*

Note on evidence: these are paraphrased, representative statements, not verbatim quotes from named sources. Before formal submission or publication, replace them with attributed review excerpts (hospital name, platform, date) or findings from primary interviews.

What published studies from Indian hospitals show

Beyond reviews and forums, peer-reviewed studies of outpatient departments (OPDs) in India point the same way: patients spend far longer waiting than they do with the doctor.

Study (setting)	Mean waiting time	Mean consultation time	Key finding
Tertiary-care hospital OPD, Gujarat (n = 80; Int J Community Med Public Health, 2021)	59.0 min	6.9 min	Shorter waits were linked to significantly higher patient satisfaction (p = 0.004).
Tertiary hospital OPDs, India (Indian J Community Med, April 2024)	98.6 min	6.5 min	General medicine averaged 122 min; paediatrics 43 min. The authors describe a clear imbalance between waiting and consultation time.
Radiation oncology OPD, Uttarakhand (J Clin Diagn Res)	about 50 min (84.1% of time in OPD)	about 10 min (15.9%)	Delays in the waiting area were the main reason for the long visit.
30 hospitals, Nellore district (n = 830; Int J Community Med Public Health, 2018)	20.3 min (government), 15.5 min (private), 39.7 min (voluntary)	Not reported	Waiting time varies strongly by hospital type, which shows that it can be improved.

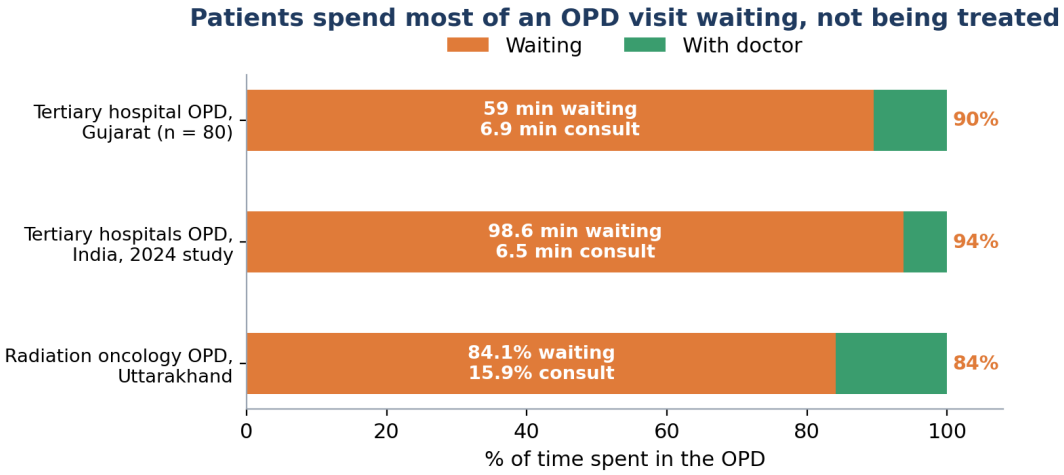


Figure 1. *Share of OPD time spent waiting vs. with the doctor, from published Indian studies (percentages calculated from the reported averages).*

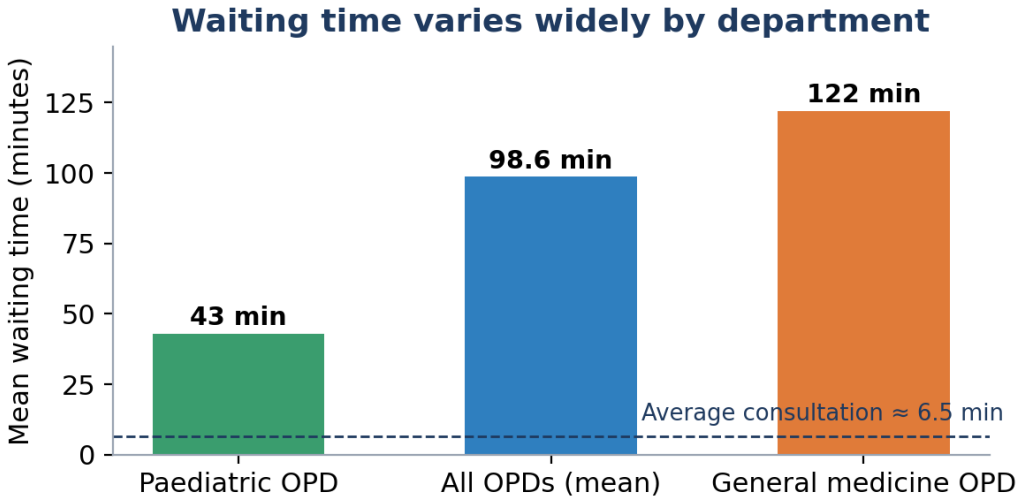


Figure 2. *Mean waiting time by department in a 2024 Indian study of tertiary-hospital OPDs.*

Two things stand out. First, the average consultation is only about 6 to 7 minutes, so a small change in queue flow has a large effect on waiting. Second, the averages hide wide variation by department and hospital, which is why a prediction that adapts to each doctor matters more than a single fixed estimate. A widely cited benchmark, from the US Institute of Medicine, is that at least 90% of patients should be seen within 30 minutes of their scheduled time.

> **Note on the "1–4 hours" claim:** published averages in tertiary OPDs are closer to 1 to 2 hours, with some patients waiting much longer. The problem statement therefore describes "an hour or more" and uses the studies above as evidence.

Digital health context: ABDM

India's Ayushman Bharat Digital Mission (ABDM) launched the ABHA-based "Scan and Share" service (now called Scan and Register) in October 2022. Patients scan a QR code at the registration counter and share their ABHA profile to get an OPD token instantly. The National Health Authority reports that it has crossed 25 crore OPD registrations across about 30,800 facilities in 756 districts, with nearly four lakh registrations a day.

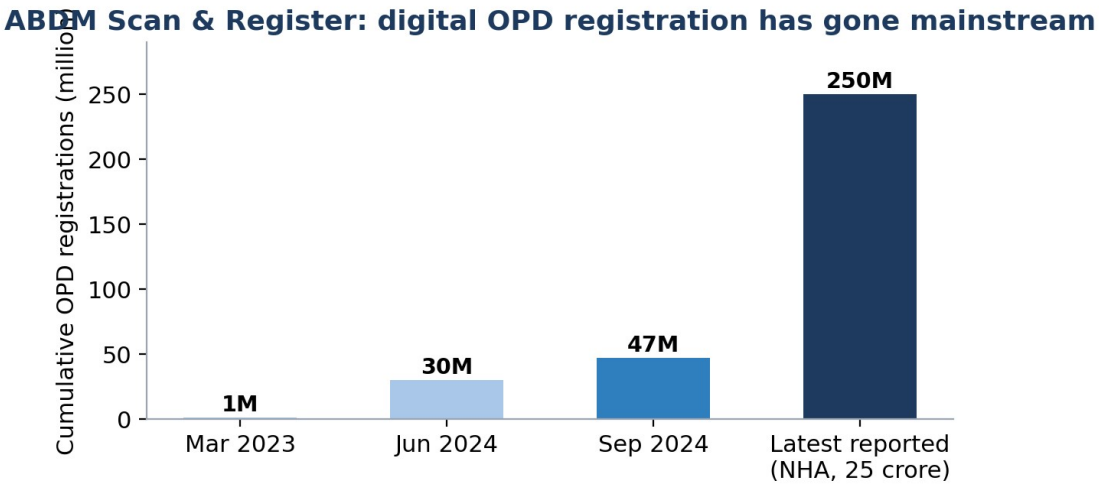


Figure 3. *Cumulative OPD registrations through ABDM Scan and Share / Scan and Register (million). The latest figure is from NHA reporting; its exact date is not stated in the source.*

This is good news for QueueLess in two ways. It shows that Indian patients and hospitals are already comfortable with QR-based digital workflows. And it exposes the gap that is still open: ABDM speeds up the registration queue, but nothing yet tells the patient how long the consultation queue will take.

Stage of the visit	Digital support today	Gap QueueLess fills
Registration and token	ABDM Scan and Register gives an instant digital token	None. QueueLess should integrate with it rather than replace it.
Waiting for the doctor	Mostly paper tokens and shouting names; no live position	Live position, estimated time and alerts.
Consultation	Doctor sees patients in token order	Data on actual consultation time to feed the prediction model.
After the visit	Records may be shared through ABHA	Feedback on waiting time to improve predictions.

Market opportunity

Healthcare digitisation in India is accelerating. Online consultations, digital health records, appointment booking and AI-assisted care are all growing, and national initiatives such as the Ayushman Bharat Digital Mission are pushing hospitals towards digital workflows. Real-time queue management, however, is still underdeveloped in most hospitals, which is the gap QueueLess targets.

3. Product Thinking

Instead of asking, “How do we build another appointment-booking app?”, we ask **“How can we eliminate unnecessary waiting time?”** This moves the focus from features to the patient’s real problem: not the wait itself, but not knowing how long it will be.

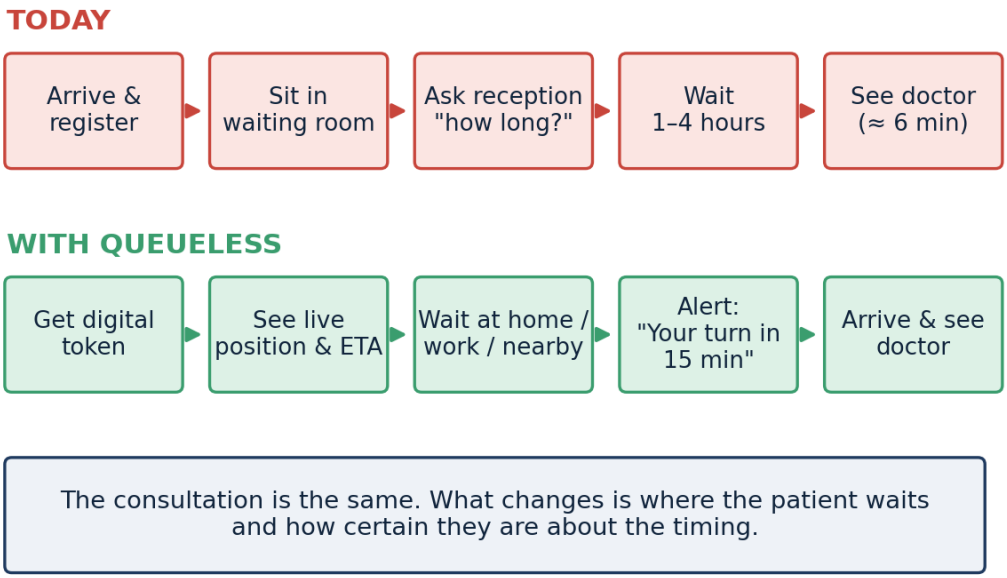


Figure 4. *The patient journey today compared with the journey with QueueLess.*

Proposed solution: QueueLess

QueueLess is an AI-powered queue prediction system for hospitals and clinics. Its core components are:

- Live queue tracking, so patients can see their position in real time
- Estimated consultation time that updates as the day unfolds
- Delay notifications by app and SMS
- A digital token system that replaces paper slips
- Smart appointment scheduling that adjusts to actual doctor pace
- AI prediction based on each doctor's consultation speed, appointment type and historical data

How the AI prediction works

Input	Why it matters
Doctor's average consultation time	Different doctors and specialities run at different speeds
Appointment type (new, follow-up, procedure)	First visits and procedures take longer than follow-ups
Historical no-show and late-arrival rates	Gaps free up time earlier than the schedule suggests
Walk-ins and emergencies logged by reception	Unscheduled cases push the queue back
Live consultation start and end times	Lets the estimate correct itself throughout the day

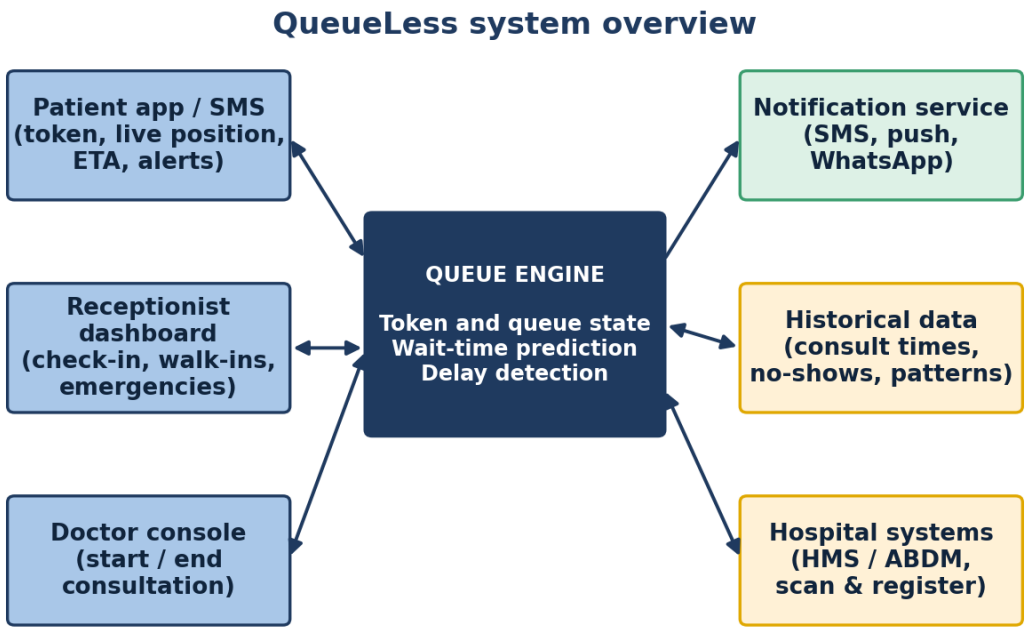


Figure 5. *High-level system overview: patient app, reception dashboard, doctor console and the queue engine behind them.*

4. Value Proposition Canvas

Customer profile

Customer jobs	Customer pains	Customer gains
Visit a doctor Get a consultation Save time	Waiting for hours Uncertainty about timing Crowded waiting rooms Missed work Stress	Faster consultation Accurate timing Less stress Better hospital experience

Value map

Products and services	Pain relievers	Gain creators
QueueLess mobile app Hospital dashboard AI queue prediction Digital token system	Live queue updates SMS notifications AI waiting-time prediction Dynamic appointment scheduling	Reduced waiting time Better hospital efficiency Higher patient satisfaction Improved doctor utilisation

Problem–solution fit

QueueLess targets the single biggest pain point: waiting without knowing when the consultation will happen. Instead of sitting in the hospital, patients receive live updates and arrive only when their turn approaches.

5. MVP Scope and Roadmap

To keep the first release focused and testable, the MVP covers only what is needed to remove the uncertainty of waiting.

Stage	Scope
MVP	Digital token; live queue position; rule-based wait estimate; SMS and app notifications; simple receptionist dashboard
Version 2	AI wait-time prediction trained on hospital data; delay alerts; dynamic slot adjustment
Version 3	Multi-doctor and multi-department support; analytics for management; integration with hospital management systems

6. Problem–Solution Fit Test

Testing method

- Build a clickable prototype in Figma.
- Recruit five hospital patients and two receptionists for moderated sessions.

Tasks participants will complete

- Book an appointment
- View queue status
- Receive and understand a delay notification
- Check the estimated consultation time

Success metrics

The concept will be considered validated if:

- At least 80% of participants understand the interface without guidance.
- At least 80% report lower anxiety about waiting.
- At least 4 out of 5 patients say they would use the app regularly.
- Hospital staff report fewer manual enquiries about waiting times.

> **Note:** with seven participants this is a directional, qualitative test. If the results are promising, follow up with a pilot in one hospital department and measure actual waiting time before and after launch.

7. Risks and Mitigations

Risk	Mitigation
Inaccurate wait predictions reduce trust	Start with conservative estimates shown as ranges; track prediction error and retrain continuously
Walk-ins and emergencies disrupt the queue	Let reception log them in one tap; show patients a “delayed” status instead of a false time
Patient health data privacy	Collect minimum data; get explicit consent; comply with India’s Digital Personal Data Protection Act, 2023
Staff do not adopt the dashboard	Keep the interface to a few taps; train reception staff; show them the drop in enquiries
Low smartphone or digital literacy	SMS and kiosk fallback so no patient is excluded

8. Success Metrics and Business Impact

North-star metric: average actual time spent in the waiting room per patient. Supporting metrics are prediction accuracy (estimated vs. actual wait), patient satisfaction score and the number of waiting-time enquiries to reception.

For patients	For hospitals
Less waiting **Better experience** **Time savings**	Improved operational efficiency Better patient satisfaction Reduced crowding Higher patient retention

Illustrative size of the prize

A simple calculation shows why even a modest improvement matters. Assume (illustratively) that QueueLess lets patients avoid about 50 minutes of in-hospital waiting per visit, roughly half of the 98.6-minute average in the 2024 study. Nearly four lakh OPD registrations are made through ABDM Scan and Register every day.

> *400,000 visits × 50 minutes = 20 million minutes, or about 333,000 patient-hours saved per day.*

This is an upper-bound illustration, not a forecast: it assumes every one of those visits is covered and saves 50 minutes. The real figure depends on how many hospitals adopt QueueLess and how well the predictions work.

Next Steps

When	Action	Output
Weeks 1–2	Interview 10 patients and 3 reception staff in one hospital; time 50 real consultations.	Real baseline for waiting and consultation times
Weeks 3–4	Build and test the Figma prototype against the success criteria above.	Validated flow and interface
Weeks 5–12	Pilot a rule-based queue in one department; collect actual start and end times.	Pilot waiting-time results and training data
Month 4 onwards	Train the AI prediction model; add ABDM Scan and Register integration.	Prediction accuracy report; go or no-go for scale-up

Conclusion

QueueLess replaces uncertain waiting with real-time visibility and AI-assisted scheduling. Rather than simply digitising appointments, it improves the whole patient experience through better information and smarter queue management. With a clear problem, a focused MVP and a practical validation plan, it represents a strong product opportunity.

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